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Occupation and Gender

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Abstract and Keywords

Occupational differences by gender remain a common feature of labor markets. This chapter begins by documenting recent trends in occupational segregation and its implications. It then reviews recent empirical research, focusing on new classes of explanations that emphasize the role of gender differences in psychological traits, preferences for nonpecuniary (family-friendly) job characteristics, personality traits, and skills. Using detailed data on occupational work content from O*NET linked to the American Community Survey (ACS), the chapter examines how the various job attributes identified in the literature affect men's and women's occupational choices and the gender wage gap. Finally, the chapter considers the role of gender identity and social norms in shaping occupational choice and preferences for various job attributes. It concludes with policy implications and suggestions for future research.

Keywords: gender, occupation, segregation, gender preferences, family-friendly, psychological traits, personality traits, identity

DESPITE the converging roles of men and women in the labor market, occupational segregation, defined as the tendency of men and women to work in different occupations, continues to be a strong feature in labor markets around the world. Gender differences in occupation and industry have been widely found to contribute to gender wage differences.¹ Recent work by Blau and Kahn (2017) highlights that as women have closed the gap in human capital accumulation, traditional human capital variables such as education and experience explain little of the gender wage gap, while gender differences in occupation and industry continue to be important. In fact, in 2010, occupation and industry were found to constitute the largest portion of the explained component of the gender wage gap. These trends highlight the importance of understanding the causes and consequences of differences in employment distributions by gender.

In this chapter, we begin by discussing the measurement of occupational segregation and trends in occupational segregation. Next, we examine the implications of occupational segregation, focusing on its impact on gender differences in pay. In the sections that follow, we review a range of potential explanations, focusing in particular on new classes of explanations for gender differences in occupational choice. Earlier work typically attributed patterns of occupational segregation by gender to differences in human capital accumulation (either pre- or postlabor market entry) or to discrimination being more pronounced in some occupations than others. While gender differences in preferences for particular job attributes have been suggested as a potential explanation for gender differences in occupational choice, empirically, it has been quite difficult to distinguish between preferences and discrimination. We review recent advances in understanding gender differences in preferences for job attributes, including research emphasizing the potential role of systematic differences in psychological traits between men and women, such as risk preferences, attitudes toward competition, and social preferences. We also consider the role of women's relative preference for nonpecuniary (p. 426) (family-friendly) job characteristics that may lead them to choose lower-paying occupations that offer better career-family balance.

We then review the recent literature that highlights the role of gender differences in personality traits and other sources of gender differences in skills that might contribute to differences in occupational choices, such as sex differences in interpersonal styles, social skills, and sensory and motor skills. To provide an overview of how these previously identified factors contribute to occupational segregation and gender pay differentials, we use detailed data on occupational work content, activities, and skills from the Occupational Information Network (O*NET) database, linked to the 2009–2011 American Community Survey (ACS), to conduct a comprehensive descriptive analysis of the relationship between the occupational characteristics discussed in the recent literature gender differences in occupational choice, and gender wage gaps. Finally, we explore the role of gender identity and social norms regarding the appropriate behavior and roles of men and women in society and how this might shape occupational choice. We conclude with some suggestions for future research and policy implications.

Measurement and Trends in Occupational Segregation

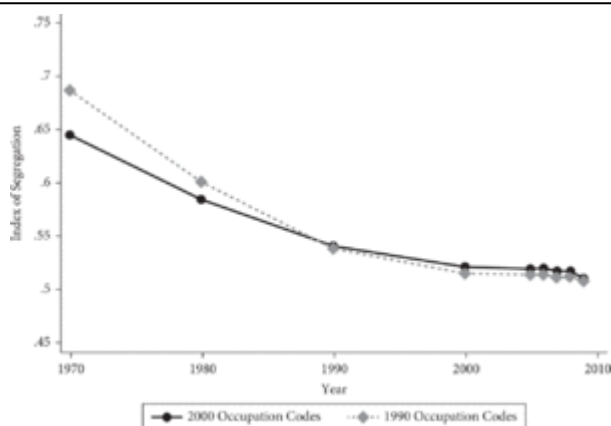
A commonly used measure to summarize differences in the distribution of women and men across occupation categories is the index of segregation developed by Duncan and Duncan (1955). The index of occupational segregation by sex is computed as

$$D = 0.5 \sum_j |M_j - F_j|$$

where M_j (F_j) is the fraction of all employed males (females) who work in occupation j . The index, which ranges between zero and one, indicates the proportion of women or men that would need to change occupations for the occupational distribution of men and women to be the same. In other words, if the distribution of men and women across occupational categories were identical (complete integration), the segregation index would equal zero. If all the occupations were either completely male or completely female (complete segregation), the segregation index would equal one.

When considering changes in the segregation index over time, it is important to distinguish between two potential channels. First, a decline in the segregation index could occur because of changes in sex composition within occupations, which occurs, for example, when women enter male-dominated occupations or, conversely, when men enter female-dominated occupations. Second, shifts in the occupation mix of the economy away from predominantly male or predominantly female occupations would cause a decline in the segregation index, even if the extent of “within occupation” segregation remained unchanged (Gibbs 1965; Blau, Brummund, and Liu 2013). Fuchs (1975) (p. 427) proposed a method to decompose the overall change in segregation into a component that captures changes in sex composition and a component that captures changes in the occupation mix.

As discussed in Blau, Simpson, and Anderson (1998), historical evidence suggests that the index of segregation in the US was relatively stable from 1900 to 1950 at about 0.66 to 0.68. Blau et al. (2013) provide a systematic analysis of the trends in occupational segregation by gender from 1970 to 2009. As observed in Figure 18.1 (reproduced based on Table 18.2 in Blau et al. 2013), there was a large decline in the segregation index over the period, with the rate of decline diminishing substantially over the decades. Between 1970 and 1980, the index declined rapidly by about six to nine percentage points. Between 1980 and 1990, the index declined by about four to six percentage points. The index continued to decline from 1990 to 2009, albeit at a diminished pace. Overall, these trends suggest that while there has been increasing gender integration across occupations over time, the degree of occupational segregation remains relatively high (according to Massey and Denton 1993, an index of segregation of 0.3 and below is considered low, between 0.4 and 0.6 is considered moderate, and above 0.6 is considered high).



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Figure 18.1 Trends in occupational segregation by gender: 1970 to 2009.

Notes: The figure is based on Table 2 from Blau, Brummund, and Liu (2013). The index of segregation is computed using census data and the gender-specific, Current Population Survey-based crosswalk using year 2000 occupational codes and year 1990 occupational codes.

Blau et al. (2013) show that the reductions in occupational segregation can be attributed to both the sex composition and occupation mix effects in each decade, with the exception of the 2000s, where the sex composition effect accounted for the bulk of (p. 428) the (modest) reduction in segregation. This decomposition highlights that the slowing decline of the segregation index has been accompanied by a corresponding decline in the magnitude of the sex

composition effect, suggesting that slowdown in sex segregation does indeed represent a slowing in gender integration within occupation, rather than changes in the occupation mix in the labor market.

Consistent with the findings from earlier studies, the authors demonstrate that the decline in the sex composition effect was largely due to the movement of women into male-dominated occupations rather than a movement of men into female-dominated occupations.² This effect was further reinforced by the significant increase in the female share of total employment from 1970 to 2009. This uneven redistribution of women led to a substantial decline in the share of men in heavily male occupations, and had less of an impact on the share of women in heavily female occupations. These changes were accompanied by a substantial increase in the share of women in moderately (60 to 80 percent) female occupations, suggesting that the entry of women into initially male-dominated or integrated occupations may cause some occupations to resegment or “tip” toward becoming predominantly female (Strober and Arnold 1987; Reskin and Roos 1990; Pan 2015).

To understand the persistence of occupational segregation, Pan (2015) investigates the dynamics of the process through which occupations have responded to the entry of women into the labor force. She documents that the dynamics of occupational segregation appear to be highly nonlinear and exhibit tipping patterns—that is, as the share of women in an occupation exceeds a critical threshold or range, men “flee” the occupation, leading to rapid feminization. These “tipping points” could arise from a Schelling (1971) social interaction model where tipping results from male preferences concerning the fraction of females in their occupation. Men’s aversion to working in occupations with too many females may be due to the existence of society-wide gender-

job associations that impinge on masculine identities (Akerlof and Kranton 2000). Goldin's (2013) "pollution" model of discrimination suggests that men's reluctance to associate with females in the workplace could also be due to informational constraints (this model is described in detail in a later section). Consistent with the predictions from the Schelling model, Pan (2015) finds that occupations tip at a lower female share in regions where men hold more sexist attitudes toward the appropriate role of women. The tipping phenomenon can provide a partial explanation for the persisting levels of segregation observed in the labor market today, and may also help explain why the historical path to gender equality has been relatively slow.

Occupations and the Gender Pay Gap

The persistence of relatively high levels of occupational segregation, coupled with large differences across occupations in pay, suggests that the underrepresentation of women in high-paying, male-dominated professions could help account for the gender pay gap. A large literature shows that female occupations pay less than male occupations with similar measured characteristics (Blau and Kahn 2017; Levanon, England, and Allison 2009).

Blau and Kahn (2017) perform a decomposition of the gender pay gap and document a rise in the relative importance of occupation and industry in explaining the gender wage gap. They find that in 1980, gender gaps in occupation and industry together accounted for 20 percent of the gender pay gap; by 2011, both factors explained more than half of the gender wage gap, and constituted the largest measured factors that account for male–female differences in wages.³ This underscores Goldin’s (2014, p. 1094) observation that “as women have increased their productivity enhancing characteristics and as they ‘look’ more like men, the human capital part of the wage difference has been squeezed out.”

While much of the literature has focused on gender-based sorting across occupations and its implications for the gender pay gap, Goldin (2014) emphasizes that the fact that a significant gender wage gap remains even after accounting for occupation and industry differences highlights the need to look *within* occupations to understand how jobs are organized and compensated and how this might differentially affect men and women. Specifically, she argues that some jobs disproportionately reward individuals who are willing to work long (and particular) hours, and that these same occupations tend to impose the largest penalties for workforce interruptions. As women tend to place a higher value on temporal flexibility, the result is a classic compensating differential equilibrium (Rosen 1986)—women and men sort across workplaces and/or job positions accordingly, and workers earn premiums for providing long hours of work in workplaces that face higher costs of providing the amenity. Using a cross-section of occupations from the 2010 ACS, Goldin (2014) provides support for this view by showing that the returns to working long hours are higher in occupations that are characterized by less flexible work schedules or requirements (e.g., meeting deadlines, contact with others, establishing and maintaining interpersonal relationships, adhering to schedules, degree to which worker has close substitutes). Moreover, she shows that occupations characterized by higher returns to long hours (nonlinear pay) are also those with larger gender gaps in earnings. To the extent that the measured returns to working long hours have increased considerably over time across education groups, and for most occupations (Kuhn and Lozano 2008; Cortes and Pan forthcoming), this might provide an explanation for the slowdown of the convergence of the gender pay gap in the past two decades (Cha and Weeden 2014).

Factors Affecting Occupational Segregation by Gender

We review a range of factors that might explain gender differences in occupational choice, focusing, in particular, on recent research that highlights the potential (p. 430) importance of gender differences in tastes and preferences for various job attributes, gender differences in personality traits, and other skill differences. To provide an overall sense of how these previously identified factors affect occupational segregation and the observed gender wage gap, we conduct a comprehensive descriptive analysis of the relationship between these factors and gender differences in occupational choice and pay using detailed data on occupational work content from the O*NET linked to individual micro-data from the American Community Survey (ACS). Finally, we discuss the potential role of gender identity and social norms in shaping occupational choice.

Gender Differences in Tastes and Preferences for Job Attributes

In this section, we discuss evidence of gender differences in tastes and preferences that might contribute to the persistence of occupational segregation and of gender gaps. We distinguish between preferences or tastes in two distinct dimensions. The first relates to gender differences in psychological attributes or preferences that may interact with the nature or content of the job to make some occupations more attractive to women and others more attractive to men—specifically, gender differences in risk preferences and attitudes toward competition, as well as social preferences (importance of social contribution vs. success and money). The second relates to preferences for certain job amenities such as temporal flexibility.⁴ These distinct preferences by gender may translate into gender differences in earnings due to compensating differentials in which women are willing to give up higher earnings to obtain other job attributes.

Attitudes toward Risk and Competition

Gender differences in risk aversion and competitiveness have been found in the laboratory and in the field (see excellent surveys by Bertrand 2011, Azmat and Petrongolo 2014 Cobb-Clark (this volume), and Schurchkov and Eckel (this volume)).⁵ As jobs in different sectors differ on a variety of attributes such as job security, earnings stability, injury and fatality risk, performance evaluation, and degree of competition, systematic gender differences in attitudes toward risk and competition could directly affect the types of jobs that men and women sort into and, consequently, gender gaps in earnings.

Despite the potentially important role that gender differences in risk aversion might play in explaining occupational segregation, the literature on the topic is scant, likely due to data availability, as most labor force surveys do not include direct measures of risk aversion. DeLeire and Levy (2004) overcome the lack of direct measures by using family composition as a proxy for risk aversion. Their hypothesis is that single parents and married mothers with children are more averse to death risk in the job. Using data from the Current Population Survey (CPS) and data on fatal and nonfatal risks associated with each occupation from the Bureau of Labor Statistics, they find that within gender, single parents are more averse to risk and tend to sort into safer occupations. (p. 431) They also find that women are much more risk averse than men, and women with no children are more risk averse than single fathers. Sloane and Grazier (2006) find similar results for the United Kingdom. DeLeire and Levy (2004) argue that differences in fatality risk across occupations can explain about a quarter of gender segregation by occupation.

Bonin et al. (2007) study the effect of earnings risk, measured as the variation in the unexplained component of wages, on occupational choice. Utilizing the German Socio-Economic Panel, which includes a self-reported measure of risk aversion, the authors find that individuals willing to take more risks earn more and sort into occupations with greater earnings dispersion. An important limitation of the study is that the sample is restricted to males; therefore, it does not directly address the role of gender differences in risk aversion on occupational segregation and gender wage gaps. Nonetheless, to the extent that women have been found to be consistently more risk averse than men, the findings from this study suggest that women are likely to end up being overrepresented in low-risk occupations that offer lower earnings variance and, as a result of compensating wage differentials, earn lower pay.

The literature on the role of gender differences in attitudes toward competition in explaining differential sorting into jobs is considerably larger. Flory, Leibbrandt, and List (2014) examine whether competitive workplaces deter female workers using a field experiment in which they randomize close to nine thousand job seekers into job ads with varying compensation schemes. They find large gender differences in the willingness to apply for a job with a more competitive compensation scheme. By varying other aspects of the job ads, the authors find that teamwork, the age of the job seekers, and whether the job position has overt gender associations attenuate gender differences in preferences toward competition. Smaller gender differences are found in studies that utilize survey data to infer preferences toward jobs that offer more competitive

compensation schemes. Using data from the British Workplace Employment Relations Survey (WERS), Manning and Saidi (2010) find that women in the United Kingdom are slightly less likely than men to be employed in jobs in which compensation is based on performance, and that gender differences in the receipt of performance pay explains only a small portion of the gender wage gap in their data. Using National Longitudinal Survey of Youth (NLSY) data, McGee, McGee, and Pan (2015) find consistent results for the United States, although their results suggest slightly larger gender differences in the incidence of performance pay and its role in explaining the gender pay gap, particularly for the most recent cohort. Nonetheless, the observed gender differences in the likelihood of working in a performance pay job do not appear anywhere as large as the analogous gender difference observed in laboratory and field studies. One reason for this difference is that in the laboratory, subjects choose between environments fully characterized by differences in competitiveness, whereas in the labor market, workers choose between jobs with many characteristics, and competitiveness may not be the most important preference when considering gender differences in the labor market (McGee et al. 2015). Furthermore, laboratory experiments may be designed in such (p. 432) a way as to magnify differences, whereas in reality, such compensation or incentive schemes may not be observed in the labor market (Manning and Saidi 2010).

Several papers have examined the role of tastes for competition in career choice. Buser, Niederle, and Oosterbeek (2014) combine an experimental measure of competitiveness with the academic track choice for a sample of high school students in the Netherlands. They find that more competitive students (according to the laboratory measure of competitiveness) were significantly more likely to choose the more prestigious academic tracks of math and science, even after controlling for ability measures. Moreover, gender differences in the laboratory measure of competitiveness explain about a fifth of the gender gap in academic track choices among Dutch students. In contrast, Reuben, Wiswall, and Zafar (2017) do not find a significant relationship between laboratory measures of competition and college major choice in a sample of students from New York University. They do, however, find that competitiveness is positively correlated with expected earnings, even within majors. They conclude that gender gaps in competitiveness and overconfidence explain about 18 percent of the gender gap in earnings expectations. Finally, Reuben, Sapienza, and Zingales (2015) combine laboratory measures of preferences toward competition with labor market outcomes for a sample of MBA students from a top business school. Their findings suggest that more competitive individuals, as measured by their choices in the lab, earn nine log points more than their less competitive counterparts, after controlling for demographic and ability measures. They show that one channel through which competitiveness affects salary is industry choice, with more competitive students more likely to go into finance and consulting. The authors estimate that gender differences in tastes for competition account for close to 10 percent of the gender gap in earnings.

One potential rationalization for women's distaste toward competition is that women may underperform in competitive settings. Several laboratory experiments have documented such gender differences in performance in response to competitive pressure (Gneezy, Niederle, and Rustichini 2003; Gneezy and Rustichini 2004).⁶ To the extent that these differences extend to the labor market, they might play a role in explaining gender pay gaps. Evidence from the field, however, is mixed. Earlier studies by Lavy (2013) and Paserman (2010) examine gender differences in the performance of high school teachers and professional tennis players, respectively, and find little evidence that women underperform in more competitive settings. More recently, a number of studies focusing on real-world academic settings show that men appear to outperform women when competitive pressures are higher, whereas the reverse holds true in less competitive settings (Cai et al. 2016; Morin 2015; Azmat, Calsamiglia, and Iriberry 2016; Örs, Palomino, and Peyrache 2013; Attali, Neeman, and Schlosser 2011; Jurajda and Munich 2011). As most of the existing studies focus on specific occupations (e.g., high school teachers and tennis players) or academic settings, the extent to which such performance differentials can explain actual gender differences in labor market outcomes remains unclear.

(p. 433) Social Preferences: The Importance of Social Contribution versus Success and Money

Women and men could select into different occupations if they differ in terms of the importance that they attach to certain job attributes, such as the perceived social value of the work in an occupation and opportunities to interact and empathize with people. Gender differences in social preferences have been studied through both experiments and surveys. In laboratory experiments, the focus has been on differences in altruism. The main findings suggest that women are not in absolute terms more socially oriented or altruistic, but that their altruism is different. Andreoni and Vesterlund (2001) find that when altruism is expensive, women are kinder, but when it is cheap, men are more altruistic. Reviewing the existing literature, Croson and Gneezy (2009) conclude that women are neither more nor less socially oriented, but their social preferences appear to be more malleable and context dependent.

More consistent gender differences in social preferences are found in surveys. Using data from the National Longitudinal Study of the High School Class of 1972 (NLS72) and the National Education Longitudinal Study of 1988/94 (NELS88), Fortin (2008) documents important gender differences in the importance attached to money/work versus people/family. In particular, she reports that the share of men who answer that "the chance to be a leader" is very important in selecting a career and that "having lots of money" is very important in life is ten percentage points larger than the share for women. In contrast, women are ten percentage points more likely than men to state that "opportunities to work with people rather than things" and "opportunities to be helpful to others or useful to society" are very important in choosing a career. Based on wage regressions that control for these preference measures, Fortin concludes that gender differences in valuing money/work and people/family account for 1.7 of the 23-log-point gender gap in

1986. This role is larger than the role of gender differences in educational attainment and cognitive skills combined and is almost as important in magnitude as the role of gender differences in labor market experience and job tenure. Grove, Hussey, and Jetter (2011) report similar findings as Fortin (2008) in their study of students who took the GMAT entrance exam for graduate business school in the United States. Similarly, Pinker (2008) argues that women tend to prefer jobs that require empathy and interacting with people, which could explain why women are overrepresented in people-oriented occupations and “care work” such as teaching and nursing (Folbre 2012). Lordan and Pischke (2016) show that women report higher levels of job satisfaction and are less likely to leave people-oriented occupations. By contrast, men are typically less affected by the degree of people-orientedness of their occupations. Consistent with the notion that men place a higher value on money, several studies have shown that expected earnings or social status is a stronger determinant of college major choices for men relative to women (Freeman and Hirsch 2008; Montmarquette, Cannings, and Mahseredjian 2002; Boudarbat and Montmarquette 2009; Zafar 2013).

What is missing from these studies is a direct exploration of the channel through which gender differences in prosocial preferences affect occupational choice and the gender wage gap. In a later section, we attempt to address this gap by using the O*NET database to construct an occupation-specific measure of social contribution and relate it to gender differences in occupational choice. (p. 434)

Preferences for Workplace Flexibility

Another dimension that might lead men and women to make different job choices is an occupation's demand for workers' time. Some occupations require employees to put in long hours of work, work at nonstandard times, or be available at short notice. As women typically have greater household responsibilities, they tend to place a higher value than men on workplace flexibility, and thus have a preference for jobs that allow a greater degree of temporal flexibility. For example, based on our calculations from the 2005 International Social Survey Programme (ISSP),⁷ if given the option, 50 percent of females aged 25 to 64 and only 25 percent of males would prefer a part-time job to a full-time job. Similarly, more women than men consider that "it is important or very important that a job allows them to decide times or days of work." Finally, even though, on average, women work fewer hours than men, they are 20 percent more likely to answer that they will "prefer working fewer hours and earning less money" than "working more hours and earning more money" or "working the same number of hours and earning the same money."

Recent work by Flabbi and Moro (2012) and Wiswall and Zafar (2018) seeks to quantify these gender differences in preferences for workplace flexibility. Flabbi and Moro (2012) estimate preference parameters for a search and matching model with wage bargaining in which both flexible and nonflexible jobs are offered. Using data from the CPS, they find that more than one-third of women place a positive value on flexibility; however, willingness to pay for it is not high. College-educated women with a positive preference are willing to pay between 1 and 10 cents per hour to work part time, whereas those with at most a high school degree are willing to pay about 2.5 cents per hour. Utilizing a survey of hypothetical job choices administered to undergraduate students from an elite university, Wiswall and Zafar (2018) estimate larger willingness to pay for workplace flexibility. They find that women have significantly higher willingness to pay for a job that offers the option of working part time than men—females are willing to give up 7.3 percent of their annual salary for a job with a part-time option; by contrast, men are willing to give up 1 percent of their annual salary.⁸

A small but growing number of papers have examined how workplace time requirements affect women's decisions to participate in the labor market and occupational choice. Pertold-Gebicka, Pertold, and Gupta (2016) document a switch from private to public sector jobs after the birth of a child in Denmark. Herr and Wolfram (2012) document that among Harvard graduates, women in flexible jobs—defined as the capacity to cut one's hours—are five to six percentage points more likely to remain working after having children. Using longitudinal data from the Survey of Income and Program Participation (SIPP), Cha (2013) shows that mothers are more likely to exit male-dominated occupations when they work fifty hours or more per week, but the same effect is not observed for men or childless women. Wasserman (2017) focuses on (p. 435) the decisions of female medical residents and finds that a reduction in weekly residency hours of medical specialties induced women to enter those specialties. Cortes and Pan (2016) focus on the broader population of college-educated women and find that an increase in

the share of males working more than fifty hours a week in an occupation reduces the share of young mothers working in that occupation, even after controlling for the occupation distribution of childless married women or men. Their findings suggest that recent increases in the prevalence of overwork might explain the persistent underrepresentation of mothers in engineering and business.

A separate but related set of recent papers explore the relationship between how occupations reward flexibility and the gender pay gap. Goldin (2014) documents that occupations vary in terms of how they reward long hours of work and that occupations characterized by a higher degree of convexity in the relationship between earnings and weekly hours are also those with the largest gender wage gaps. Cha and Weeden (2014) document that rising returns to overwork, coupled with the gender gap in the propensity to work overtime, worked to slow the convergence of the gender wage gap between 1979 and 2009. Cortes and Pan (forthcoming) find that exogenous shocks to the supply of long hours by high-skilled women—generated by intercity variation in low-skilled immigrant flows—reduce gender pay gaps among the college educated in occupations that disproportionately reward longer hours of work.

The role of the demand and supply for temporal flexibility in the gender composition of occupations is nicely exemplified in Goldin and Katz's (2016) work on pharmacists. They document how increased substitution between pharmacists (resulting from technological changes) and the growth of retail chains at the expense of independent pharmacies reduced the prevalence and returns to working long hours attracting women to the occupation. Today, about 55 percent of pharmacists are female compared to 18 percent in 1980, and the gender earnings gap is much smaller than in almost any other college graduate field.

Gender Differences in Personality Traits and Skills

Traditional supply-side explanations for gender differences in job choices and gender wage gaps tended to focus on differences in human capital accumulation between men and women (Altonji and Blank 1999). As the gender education gap has reversed in favor of women, recent studies have considered a more nuanced view of gender differences in personality traits and skills and how these skills are rewarded in the labor market. These differences in personality traits and skills can be regarded as part of an individual's set of productive traits that are directly valued in the market, thus affecting how men and women sort into different occupations or jobs and earnings (Bertrand 2011).

An obvious difference between men and women is that women tend to have a comparative advantage in cognitive relative to manual or motor skills, and these gender differences in “brains” versus “brawn” are likely to explain why women tend to select into occupations that are more cognitive-task intensive (Welch 2000). Several (p. 436) papers provide evidence that women have better interpersonal or “people” skills than men and are matched to occupations in which “people” tasks are more important.⁹ Borghans, ter

Weel, and Weinberg (2008) show that personality at age 16 is a significant predictor of later job assignment, and relatively caring people are more likely to work in caring jobs. Moreover, Borghans, ter Weel, and Weinberg (2014) document that the relative employment of women is higher in occupations that place a greater emphasis on people tasks, consistent with the idea that women are more well endowed in interpersonal skills. A recent paper by Baker and Cornelson (2016) investigates the extent to which sex differences in sensory, motor, and spatial aptitudes can explain the observed degree of occupational segregation. The results indicate that these skills, as captured by Dictionary of Occupational Titles (DOT) codes, strongly predict men's and women's occupational choices in the directions indicated by scientific research. The authors estimate that eliminating selection on these aptitudes would reduce the index of segregation by 23 percent in both 1970 and 2012. Moreover, accounting for gender differences in other skills such as cognitive skills, physical strength, and people skills, in addition to sensory, motor, and spatial aptitudes, would reduce the index by about one-third in both years. Using a similar approach and proxies for job skill requirements from the O*NET database, Levanon and Grusky (2012) find that physical and interactional social skills are particularly important in explaining gender occupational segregation.

Several papers focus on gender differences in interactive, analytic, and social skills to understand the impact of technological change on occupational skill requirements and female relative wages. These papers highlight that gender-based occupational selection on the basis of some personality traits and skills may increasingly favor women due to the growing importance of these skills in wage determination. In particular, these papers posit that technological changes are likely to have disproportionately benefited women as women tend to be relatively more endowed in the skills that cannot be easily automated. Using a task-based framework introduced by Autor, Levy, and Murnane (2003) and data from West Germany, Black and Spitz-Oener (2010) find that women have experienced relative increases in nonroutine analytic tasks and nonroutine interactive tasks, which are associated with higher skill levels. Moreover, women have experienced a pronounced decline in routine task inputs, in contrast to men, who experienced little change. The authors find that these relative task changes can account for a large fraction of the closing of the gender wage gap between 1979 and 1999.

Borghans et al. (2014) examine the effect of computerization on the relative demand for people/social skills and its implications for the labor market outcomes of women. They present evidence that the spread of computers appears to have increased the relative demand for interpersonal interactions, which can potentially explain the rise in women's relative wages from the late 1970s to the early 1990s.¹⁰ The authors argue that the slowing convergence of the gender-wage gap since the mid-1990s appears consistent with a slowdown in the rate of growth of the importance of people tasks, perhaps due to computers becoming more effective at substituting for interactive tasks.

(p. 437) **How Do Occupational Characteristics Affect Occupational Segregation and Gender Wage Gaps?**

Most of the studies that we reviewed in the previous sections consider how gender differences in tastes or skill for a particular occupational attribute may lead to gender differences in labor market outcomes. As these studies use different settings (laboratory vs. field vs. surveys), datasets, and outcomes, it is not easy to compare the results across studies to get an overall sense of how gender differences in preferences/tastes for the various occupational attributes affect occupational segregation and observed gender wage gaps.

In this section, we attempt to address this gap by utilizing detailed information on the work content, activities, abilities, and skills required in an occupation from the O*NET database linked to data from the 2011 ACS to provide a comprehensive descriptive analysis of how gender differences in work-related preferences and skills affect men's and women's occupational choices and the gender wage gap.

We create six composite variables using the O*NET database to describe the characteristics of an occupation, focusing in particular on the aspects of the job that map most closely to the job attributes discussed in the previous sections. Unless otherwise noted, the O*NET questions used are based on a five-point scale.¹¹ We construct the following measures:

- 1. *Competition*:** Based on the question "How competitive is your current job?"
- 2. *Social Contribution*:** Based on three questions: (a) "How important is concern for others to the performance of your current job?" (b) "How important is assisting and caring for others to the performance of your current job?" (c) "How important is service orientation to the performance of your current job?" (actively looking for ways to help people)
- 3. *Inflexibility*:** Based on two questions: (a) "How often does your current job require you to meet strict deadlines?" (1: never, 2: once a year or more but not every month, 3: once a month or more but not every week, 4: once a week or more but not every day, 5: every day); and (b) "How many hours do you work in a typical week on your current job" (1: less than forty hours, 2: forty hours, 3: more than forty hours).
- 4. *Interactional Skills*:** Based on four questions: (a) "How much contact with others (by telephone, face to face, or otherwise) is required to performance your current job?" (b) "How important are interactions that require you to work with or contribute to a work group or team to perform your current job?" (c) "How important is establishing and maintaining interpersonal relationships to the performance of your current job?" (d) "How important is social perceptiveness to the performance of your current job?"

5. Cognitive Skills: Based on the importance placed on the following four abilities for the performance of the job: (a) written comprehension, (b) mathematical reasoning ability, (c) deductive reasoning, and (d) inductive reasoning

(p. 438) **6. Physical Skills:** Based on the importance placed on the following two activities for the performance of the job: (a) general physical activities and (b) handling and moving objects

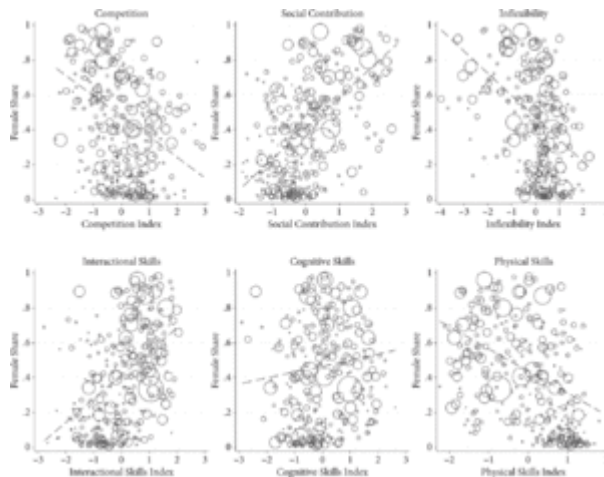
To link the O*NET measures to the occupations in the census, we use the crosswalks developed by Autor and Acemoglu (2011) and Dorn (2009). Our sample is composed of 318 occupations. The composite measures are constructed by first normalizing each of the O*NET characteristics to have a mean of zero and standard deviation of one in the full sample of occupations and taking the average of the normalized components. For ease of interpretation, we normalize the composite measures to have a mean of zero and standard deviation of one in the full, unweighted sample of occupations.

To examine the relationship between the various occupational characteristics and gender differences in occupational choice, we relate each of the six O*NET composite measures to the share of females aged 25 to 54 in an occupation in 2011. Figure 18.2 graphically depicts the cross-occupation correlations. The size of the circle represents the employment size of each occupation, and the dashed line in each panel is the fitted regression line, weighted by the employment size of each occupation. The correlations illustrated in the figure are all in the expected direction—the share of females is higher in occupations that are less competitive, place a greater emphasis on social contributions, are more flexible, require more interactional skills, and require less physical skills. There is a slight positive relationship between the fraction of females in an occupation and the measure of cognitive skills, but the correlation is not statistically significant.

Columns (1) to (6) of Table 18.1 confirm the graphical correlations presented in Figure 18.2. With the exception of the cognitive skill index, the correlations are all highly statistically significant. In column (7), we include all six composite measures together into the model to examine the independent effect of each occupational characteristic, conditional on the other occupational attributes. The magnitude of the coefficient estimate on interactional skills is now close to zero and not statistically significant, most likely because the measure of interactional skills and social contribution are highly correlated.¹² The coefficient estimates on the measures of competition, social contribution, flexibility, and physical skills remain statistically and economically significant. As shown in column (8), these results continue to hold even when we include additional controls for occupation-level characteristics such as the educational composition, average age, and racial composition of workers in an occupation. The magnitude of the estimates from the full regression specification in column (8) imply that a one-standard-deviation increase in the competition measure is associated with a 6.8-percentage-point decrease in the female share in an occupation. A one-standard-deviation increase in the social contribution index is associated with a 12.7-percentage-point increase in female share, while the corresponding increase in the inflexibility index is associated with a 5.8-percentage-point decrease in female share. Finally, a one-standard-

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deviation increase in the importance of physical skills in an occupation is associated with a 16.6-percentage-point decline in the female share in an occupation. Overall, these results offer some empirical (p. 439) evidence in support of the notion that gender differences in preferences and skills for different job attributes contribute to gender differentials in occupational choice.



[Click to view larger](#)

Figure 18.2 Relationship between various occupational characteristics and the share of females in each occupation.

Notes: The composite indices are constructed using data from the Occupational Information Network (O*NET). The female share in an occupation is computed from the 2009–2011 American Community Survey (ACS) and is based on employed workers between the ages of 25 and 54. The sample includes 318 occupations. The composite indexes have been normalized to have a mean of zero and standard deviation of one in the full unweighted sample of occupations. The size of the circles indicates the relative employment size of each of the occupations. The dashed line is a fitted line based on a weighted regression of female share on each of the occupational characteristics, using employment size as weights.

Next, we examine whether gender differences in occupational choice on the basis of these occupational attributes translate into gender differences in wages. Table 18.2 reports the gender difference in hourly wages among full-time, full-year workers in specifications with and without the various composite measures. Column (1) indicates that the hourly wages of females are about 22 percent lower than those of males after controlling for individual characteristics such as race, education, age, and marital status.¹³ Columns (2) to (6) include each of the composite measures in turn. The estimates in column (2) indicate that the underrepresentation of

women in competitive occupations lowered their relative wages, such that the competition index alone can explain about 23 percent of the residual gender wage gap. Gender differences in sorting on (p. 441) (p. 440) (p. 442) the basis of social contribution and flexibility account for about 5 percent and 27 percent of the residual gender wage gap, respectively (see columns [3] and [4]). In contrast, gender differences in occupational choice on the basis of the three skill indexes (interactional, cognitive, and physical) tended to *raise* women's relative wages. This occurs because women tend to sort into occupations with higher interactional and cognitive skills, and these skills are associated with higher wages. Moreover, women sort away from occupations that require more physical skills, which tend to offer lower wages. In column (8), we estimate a full

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specification that includes all the occupational measures and find that, together, the six occupational characteristics can explain about 14 percent of the residual gender wage gap.

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Table 18.1 Relationship between Various Occupational Characteristics and the Share of Females in an Occupation

			Dependent Variable: Female Share in an Occupation					
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Occupational Characteristics</i>								
Competition Index	-0.118** [0.028]						-0.076** [0.021]	-0.068** [0.020]
Social Contribution Index		0.176** [0.023]					0.144** [0.031]	0.127** [0.032]
Inflexibility Index			-0.127** [0.019]				-0.068** [0.015]	-0.058** [0.016]
Interactional Skills Index				0.136** [0.030]			-0.003 [0.032]	-0.000 [0.031]

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Cognitive Skills Index					0.035 [0.030]		-0.047 [0.027]	-0.043 [0.031]
Physical Skills Index						-0.128** [0.027]	-0.168** [0.023]	-0.166** [0.026]
Additional Controls for Share with High School Degree, Some College, and College; Average Age; Share of Whites, Blacks, and Hispanics	No	No	No	No	No	No	No	Yes
Observations	318	318	318	318	318	318	318	318
R-Squared	0.152	0.312	0.212	0.171	0.014	0.175	0.674	0.698

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Notes: The unit of observation is an occupation. The data is from the Occupational Information Network (ONET) and the 2009–2011 three-year aggregate American Community Survey (ACS). The ACS sample is restricted to employed workers between the ages of 25 and 54. See text for details on how the various composite indexes are constructed. The composite indexes are normalized to have a mean of zero and standard deviation of one in the unweighted sample of occupations. All regressions are weighted by the employment size in each occupation. Robust standard errors are reported in brackets.

(***) Significant at 1% level,

**5%,

(*) 10%.

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Table 18.2 Occupational Characteristics and the Gender Wage Gap

			Dependent Variable: Log Hourly Wage					
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Female	-0.224** [0.019]	-0.172** [0.017]	-0.213** [0.018]	-0.163** [0.017]	-0.253** [0.019]	-0.250** [0.013]	-0.275** [0.020]	-0.193** [0.013]
Competition Index		0.126** [0.016]						0.051** [0.016]
Social Contribution Index			-0.021 [0.023]					-0.029 [0.026]
Inflexibility Index				0.131** [0.015]				0.050** [0.015]
Interactional Skills Index					0.076** [0.024]			0.002 [0.027]
Cognitive Skills Index						0.200** [0.014]		0.151** [0.021]

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Physical Skills Index							-0.114** [0.017]	-0.019 [0.016]
Controls for Individual Characteristics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
% of Gender Gap Explained		23.2%	4.9%	27.2%	-12.9%	-11.6%	-22.8%	13.8%
Observations	1,862,951	1,862,951	1,862,951	1,862,951	1,862,951	1,862,951	1,862,951	1,862,951
R-Squared	0.299	0.330	0.300	0.335	0.307	0.359	0.321	0.375

Notes: The data is from the 2009–2011 three-year aggregate American Community Survey (ACS). The sample is restricted to individuals between the ages 25 and 54 who report working full time (thirty-five hours or more per week), full year (forty weeks or more per year). The unit of observation is an individual. The occupational indices are constructed using data from the Occupational Information Network (ONET) and are normalized to have a mean of zero and standard deviation of one in the unweighted sample of occupations. The individual-level controls include race dummies, a quadratic in age, education dummies (high school degree, some college, college), dummies for marital status (married and single), and number of children. Standard errors clustered at the occupation level (318 occupations) are reported in brackets.

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(***) Significant at 1% level,

**5%,

(*) 10%.

Gender Identity and Social Norms

Another potential explanation for the persistence of occupational segregation is the presence of slow-moving gender identity norms regarding what is the “appropriate” work for men and women. In their influential paper, Akerlof and Kranton (2000) define identity as one’s sense of self or one’s sense of belonging to a social category that comes packaged with a view about how people in that category should behave. Akerlof and Kranton (2000) propose a model where one’s identity directly enters the utility function—in this model, social identity influences economic outcomes because deviating from the prescribed behavior associated with one’s social category is costly.

Akerlof and Kranton (2000) apply the model to the notion of gender identity and consider how identity considerations may influence occupational segregation by gender. In this application, the two relevant social categories are those of a “man” and a “woman,” and these two categories entail specific behavioral prescriptions. Akerlof and Kranton (2000) cite examples of occupations that are “gendered”; for example, marines and trial lawyers are viewed as masculine, while nurses are viewed as feminine. As such, women (men) in “masculine (feminine)” occupations may incur a disutility as their actions are in conflict with the behavioral prescription for that gender category. This could explain why women may be reluctant to enter male professions. At the same time, a woman working in a “man’s” job may threaten men’s masculinity and their gender identity, prompting them to act against female coworkers, which creates a further disincentive for women to enter the occupation. In this context, the identity model provides a micro-foundation for taste discrimination models (Becker 1971), which assume that some groups have a “distaste” for working with members of another group. Here, men are reluctant to associate with women on the job because of the perceived loss in male identity, even if in other circumstances, men get along well with women.

Goldin (2014) proposes a related model that shows how identity considerations can result in men wanting to avoid working with women in the same occupation. In her pollution theory of discrimination, Goldin (2014) assumes that in addition to wages, men also value how society construes the status or prestige of the occupation. In the model, prestige is based on the level of a productivity-related characteristic, such as (p. 443) strength, skill, education, or ability, that might originally define the minimum qualification needed to enter a particular occupation. In the presence of uncertainty about changes in technology, society infers the change from observables, and female entrants into an occupation may signal that the job has undergone a negative productivity shock, even when it has not. Men therefore want to avoid having females in their occupations to protect their occupational status. In contrast to the identity model, where occupational desegregation occurs largely through change in social norms regarding gender identity, the “pollution” model emphasizes the role of imperfect information about the true productivity of female entrants into an occupation. In particular, Goldin (2013) suggests that occupational desegregation in the 1970s and 1980s could, in part, be explained by

the credentialization of occupations through the widespread use of degrees and licenses, which served to provide objective measures of workers' productivity and eliminate the negative signal of hiring women.

While the gender identity model provides a useful way for understanding how social norms and discrimination may perpetuate occupational segregation, direct empirical tests of the relevance of the gender identity model in explaining occupational segregation are few. Several recent papers provide some indirect evidence consistent with the predictions of the gender identity model. Usui (2008) finds that women report less satisfaction with job-related amenities in male-dominated jobs, while males are either indifferent or report higher satisfaction in predominantly male jobs. Using data from the United States, Britain, and Russia, Lordan and Pischke (2016) replicate these findings on job satisfaction, and additionally show that in occupations with a higher share of males, women are more likely to leave, while males are more likely to stay. They find that part of this effect for females is driven by differences in job content and context as proxied for by occupational characteristics that they label "people," "brains," and "brawn." The authors interpret these results as suggesting that women and men have different tastes for the content of the work that they do; nonetheless, the gendered nature of these occupation characteristics (e.g., female job satisfaction is lower in occupations with higher "brawn" content, but higher in occupations that have higher "people" and "brain" content) suggests that identity considerations may lead men and women to gravitate toward occupations that require the skills that match prevailing gender stereotypes.

The difficulty in empirically distinguishing whether underlying gender differences in tastes and preferences for certain occupational attributes (such as those discussed previously) are innate or stem from socially constructed gender identity norms has been discussed in some detail in the review article by Bertrand (2011). Bertrand (2011) provides an overview of studies in psychology that suggests that psychological attributes such as risk preferences, confidence, and assertiveness may be part of what people associate with the male identity, while women are expected to be docile, generous, and risk-averse (Eagly 1987; Eckel and Grossman 2002). Women's preferences for temporal flexibility may also reflect adherence to behavioral prescriptions that indicate that men should take on the role of breadwinners and women as homemakers. As such, gender differences in personality traits and preferences could be a reflection of women (and men) conforming to societal expectations of the appropriate behavior (p. 444) or role of women (and men) as dictated by the identity considerations. There is some evidence that men and women who identify with more "masculine" traits¹⁴ appear more likely to enter male-dominated fields of study and occupations (Antecol and Cobb-Clark 2013).

Two recent studies utilize a laboratory setting to establish a causal link between gender identity and preferences. Cadsby, Servatka, and Song (2013) examine how making salient gender and family identities versus professional identities may influence preferences for competition. The authors generate exogenous variation in the salience of various social identities by "priming" MBA-student participants through questionnaires concerning either gender/family or professional issues. The authors find that females primed with the

gender/family identity are significantly less competitive than those primed with the professional identity. Interestingly, opposite patterns were obtained for male subjects—males primed with the gender/family identity were significantly more competitive than those primed with the professional identity. Extrapolating these results, the authors hypothesize that one potential reason women opt out of competitive professional occupations and careers may be the increased salience of the gender/family identity over the lifecycle. Focusing instead on risk preferences, D'Acunto (2015) uses a field setting—the Amazon Mechanical Turk (mTurk) online platform—to examine how subjects' risk tolerance changes after an experimental manipulation of their gender identity. Gender identity was made salient to treated subjects by asking them to read and respond (in the form of a short essay) to a text about gender stereotypes. Men whose identities were made more salient (exposed to male-identity prime) or threatened (exposed to female-identity prime) were found to be more risk tolerant after the experimental manipulation, and tended to invest more in risky opportunities. There were no discernible effects on women.¹⁵ The author also finds that the effects appear weaker for younger cohorts of men, suggesting that gender identity stereotypes may have weakened over time.

Conclusion

Despite the remarkable progress that women have made in the labor market over the past century, occupations continue to be segregated along gender lines. The differential sorting of men and women into different occupations could be an important channel through which gender wage differences are maintained. At the same time, the same factors that lead men and women to choose different occupations may also have a direct effect on the gender wage gap. Over the last decade, labor economists have increasingly moved beyond the traditional focus on human capital differences and discrimination to explain gender differences in labor market outcomes. We reviewed three broad classes of explanations that have been the subject of much of the recent empirical work. These new classes of explanations have a lot of potential to improve our understanding of the underlying causes of gender differences in occupations and how occupation and gender (p. 445) interact to affect gender wage gaps. Nonetheless, there is clear scope for future work to address the gaps in the current literature.

While gender differences in behavioral traits such as risk aversion and attitudes toward competition are quite well established in the laboratory and field, and have been shown to matter for labor market outcomes in some settings, more work needs to be done to assess the quantitative relevance of these factors for observed gender differences in occupational choice and wages. Studies that use observational data to link measured preferences, job attributes, and occupation and wage outcomes offer some promise in addressing this question. Nonetheless, such regression-based studies (including the descriptive analysis presented in this chapter) may be susceptible to issues relating to the endogeneity of preferences for certain job attributes. Moreover, to the extent that the

share of women in an occupation might affect the workplace environment and measured occupational characteristics, there is a concern of reverse causality. Furthermore, it is challenging to attach a causal interpretation to the various behavioral/personality traits or occupational attributes included in this model, since there is a question about precisely what it is that these variables are capturing, and whether the observed relationships may be capturing omitted factors. Therefore, such evidence based on survey data may need to be combined with well-designed experiments in the field and/or plausible sources of exogenous variation to credibly identify how gender differences in preferences and behavior shape labor market outcomes.

Another aspect that would benefit from future research is an investigation of the sources of gender differences in behavioral traits and preferences. The extent to which the observed gender differences in tastes and preferences for particular job attributes are due to biological differences between men and women (nature) or are shaped by the social environment (nurture) remains an open question. The potential role of nurture in shaping these preferences and labor market outcomes is particularly important for thinking about policy responses. Risk preferences, attitudes toward competition, and social preferences may be components of one's gender identity, and the strength of these preferences may be influenced by the presence of social norms that dictate what the appropriate behavior of women (and men) should be (Bertrand 2011). Furthermore, gender differences in preferences for workplace flexibility are likely to have their roots in the traditional gender division of labor in the household, which in turn could be maintained by an adherence to gender identity norms. To the extent that gender identity and social norms may be an important underlying cause, it would be important to understand what drives these norms. Several papers have evaluated historical factors that may have affected these norms, such as the advent of the pill (Goldin and Katz 2002) and World War II (Fernandez, Fogli, and Olivetti 2004). Additional research on the role of social factors in shaping gender identity norms can help further our understanding on viable policy initiatives to address the remaining labor market gaps.

Finally, to the extent that gender differences in household roles remain pervasive, occupational differences in the degree of workplace flexibility are likely to have a differential effect on men and women. Moreover, the secular increase in workplace time (p. 446) demands over time in the United States (Kuhn and Lozano 2008; Cortes and Pan 2016, 2016, forthcoming) suggests that this factor is likely to become an increasingly binding constraint for women seeking to effectively combine career and family. What remains less well understood is why some occupations have a larger demand for long (and particular) work hours than others, and why the demand for workers to put in long (and particular) hours has increased differentially across occupations and countries. Understanding the relative importance of the underlying causes of these changes (e.g., changes in compensation schemes, technological change, market structure, and globalization) and how they may have influenced organizational practices and workplace cultures can help to facilitate the design of policies. At the same time, a careful

exploration of how occupations have successfully reorganized their workplace environment to accommodate greater flexibility could provide valuable lessons for ways to enhance temporal flexibility in occupations (Goldin 2014).

Our review of the recent literature offers a number of policy implications for tackling occupational segregation and gender pay differentials. The importance of social norms and gender identity considerations in shaping occupational choice (perhaps by affecting men's and women's tastes and preferences for certain occupational attributes) suggests that policies aimed at removing gender stereotypes and changing gender norms could have potentially large payoffs. Such policies could include exposing more women to traditionally male subjects (e.g., science, technology, engineering, and mathematics [STEM] fields) early on in school, efforts to encourage the hiring of women in male-dominated professions, and incentives and policies to retain women in these fields.¹⁶ Moreover, given gender differences in household responsibilities, policies that increase the availability of household substitutes (e.g., immigrant domestic help and affordable high-quality child care) and promote workplace flexibility can help to relax some of the constraints that women face in seeking to pursue careers in occupations that require a high degree of time commitment.

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Notes:

(1.) For example, see Groshen (1991); Macpherson and Hirsch (1995); Altonji and Blank (1999); Blau, Ferber, and Winkler (2009); and Blau and Kahn (2017).

(2.) See, for example, Cotter et al. (1995) and Blau et al. (1998).

(3.) Occupation (Industry) fixed effects explain 5.1 (4.6) and 7.6 (4.1) percentage points of the gender wage gap in 1980 and 2010, respectively. While occupation and industry explain a slightly larger percentage point wage gap (11.7 vs. 9.7) in 2010 versus 1980, the large increase in the share of the gender wage gap explained by both variables is largely driven by the decline in the total gender wage gap (47.7 percent vs. 23.1 percent) from 1980 to 2010 due, in part, to the disappearance of the human capital (education and experience) gap.

(4.) We note, however, that gender differences in both dimensions might share a common cause, namely, gender identity and social norms regarding the appropriate role or behavior of women in society.

(5.) Laboratory experiments typically use randomization to test hypotheses in the highly controlled, albeit artificial, setting of a laboratory. Field experiments apply similar experimental methods as those used in the laboratory but in the real world. Field experiments often have the advantage that the outcomes are observed in a natural setting and are therefore likely to have higher external validity than laboratory experiments. Moreover, some phenomena (e.g., occupational choice or educational choice) cannot be easily studied in a laboratory. However, the trade-off is that field experiments tend to be a lot more expensive and researchers typically have less precise control over experimental conditions.

(6.) By contrast, Niederle and Vesterlund (2007) find that men's and women's performance increased similarly in the tournament scheme relative to the noncompetitive compensation scheme.

(7.) The ISSP is a cross-national collaboration on social surveys and has data for thirty-two countries, the United States included. Each year the survey includes a module on a specific topic. The 2005 module was on "Work Orientations." The sample size for 2005 is close to forty-five thousand.

(8.) See Kunze, this volume, for a full discussion of gender wage issues.

(9.) Several studies have also shown that females score higher on tests of emotional and social intelligence (Hall 1978; Woolley et al. 2010; Kirkland et al. 2013).

(10.) Bacolod and Blum (2010) also demonstrate that increased demand for social skills might account for the narrowing of the gender wage gap.

(11.) Please refer to the O*NET data documentation for more details on the variable definitions.

(12.) In fact, when the social contribution index is omitted from the model, the interactional skills index is positive and highly statistically significant.

(13.) The gender wage gap conditional on individual characteristics is commonly referred to as the “residual gender wage gap.”

(14.) In their study, masculine traits include a set of attributes such as independence, assertiveness, (not) shy, (not) sensitive, and (not) emotional. These attributes have been identified as masculine in the Bern Sex Role Inventory (BSRI), an instrument used by psychologists to judge the masculinity or femininity of a person (Bern 1974).

(15.) An earlier study by Benjamin, Choi, and Strickland (2010) finds no effect of gender salience on risk preferences, although they uncover effects for racial salience. There are two main differences between the studies. First, Benjamin et al. (2010) use a between-subjects experimental design, while D’Acunto (2015) uses a within-subject design. Second, the experimental prime used in D’Acunto’s (2015) study was a lot more aggressive than the primes used in the Benjamin et al. (2010) study.

(16.) For a full discussion of STEM and gender issues, see Kahn and Ginther (this volume).

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